



Major Choosing Decision Support System With Talent Interest Approach Using SWRL and SAW Method

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ABSTRACT

Students who are majoring in college that are not following their interests will have difficulty in adapting. Psychologically, to learn things that are not by your interests and abilities will lead to emotion blockage, which then blocks the effectiveness of brain performance. This issue will have an impact on the inhibition of learning motivation and achievement of academic performance. To avoid these problems, a diagnosis of career interests, including interests and needs to be done as an initial process of further study planning interventions to support career decisions making in individuals. Therefore, this research builds a Decision Support System with the Talent Interest Test approach to help users find college majors that suit their interests and abilities. This study uses the SAW method and SWRL reasoning. The results of this study indicate that the implementation of this approach in a Department Selection Decision Support System with Talent Interest Approach can provide an alternative that is in line with the interest of undergraduate student candidates with the compatibility level of Agree (95.31%) and Disagree (4.69%).

CCS CONCEPTS

• Mathematics of computing~Probability and statistics~Statistical paradigms~Time series analysis • Computing methodologies~Artificial intelligence~Knowledge representation and reasoning~Vagueness and fuzzy logic • Information systems~Information systems applications~Decision support systems~Expert systems

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KEYWORDS

Career Type, Decision Support System, Major Choosing, Ontology, SAW, SWRL

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1 Introduction

The mistake in choosing a field of study at the level of higher education and higher education can cause quite serious problems [21]. Students who are studying in majors which is not following their interests will experience conflicts in the adaptation process in university [6]. According to [18], someone who is forced to learn things that are not following their interests, talents, and abilities will experience psychological problems in the form of emotional blocks, which then block the effectiveness of brain performance and inhibit motivation. A work cited in [18] also explained that a person who has no interest and ability in the department he chooses would experience learning motivation problems. Meanwhile, one's learning motivation tends to be directly proportional to the academic performance that can be achieved [7].

An Educational Psychologist from Integrity Development Flexibility quoted from [5] stated that 87% of students in Indonesia felt they were in the wrong department. A release from the Indonesia Career Center Network survey in 2017, quoted from [2], also mentioned that 87% of Indonesian students admitted that the department taken was not following their interests. Other data released by Youth Manual said 92% of high school/vocational students were even confused and did not know the plans [12].

The diagnosis of career interests, including the interests and talents of a person, needs to be done as an initial process in planning career development and intervention, both advanced studies and the profession [4]. Such assessments can be used to support career decision-making in individuals [9] and make the career decision-making process is easier to do [10]. One of the career assessment tools that has been established and used for more than 20 years is an assessment developed by Holland [3,13,14]. This tool is

widely used throughout the world because its use is easy and empirically tested [10]. In addition to diagnosing interest in talent, the interests of prospective students in the material to be studied and carried out in lectures is also a crucial factor in determining college majors [9].

Previously, several studies have been conducted related to the Major Choosing Decision Support System using the results of the aptitude interest test, which is then referred by the author as a talent interest approach. Work in 2018 used the aptitude interest test as one of the criteria in decision making in choosing a tertiary department. This study applies the Fuzzy Inference System Mamdani method and produces a system that can provide alternative majors based on the results of tests that have been done [19]. Pretty similar work was done as well [1,11].

From some of the research above, there is no research decision support system for the selection of departments using the aptitude interest test approach that applies the Semantic Web Rule Language (SWRL) in the decision making process. Meanwhile, some previous research has proven that reasoning with SWRL can help the decision-making process in a Decision Support System. Among them is research on the Decision Support System to select manufacturing processes that have shown that rule-based reasoning using SWRL allows being implemented in the decision making process [8]. Another Decision Support System research to assist the process of labeling environmentally friendly products also show how a decision support system with an SWRL-based mechanism can produce explanations and arguments by eco-labeling criteria [22]. SWRL technology on this kind of ontology-based system is proven to be used in decision support systems for various problem domains [17]. Therefore, the Decision Support System for the selection of this department implements SWRL to assist the decision-making process.

The method used in this study is the Semantic Web Rule Language and the Simple Additive Weighting (SAW) method. Several previous Decision Support System studies have used the SAW method. Research on the problem domain mapping of flood-prone areas in Semarang City with the criteria of rainfall, topography, drainage, and land use succeeded in providing information on the level of flood vulnerability [16]. Then other research to consider employee salary increases to increase company productivity also implements the SAW method and can produce a percentage increase in salary for each employee [15]. From some of these applications, it can be seen that the characteristics of this method are that it allows users to have full subjectivity in determining the weighting of criteria for decision making.

In this study, the SAW method is used for weighting criteria and determining alternatives on several criteria whereas, SWRL is used to make final decisions in the form of slices of alternative departments each of the following criteria with the value of each alternative.

2 Related Work

Several studies related to the Decision Support System for Department Selection with the Talent Interest Test Approach have been

conducted previously. A work [11] uses the Multifactor Evaluation method and produces a Decision Support System that can provide department recommendations that are by the interests of prospective students. Further research in 2018 applies the Dempster-Shafer method in the Decision Support System for the selection of departments. It also uses the criteria for the Holland Test. The results of this study get a good response from prospective students [1]. The weakness in these studies is that the aptitude interest test used is not available in the system.

Some of the Decision Support System research that has been done applies the Simple Additive Weighting (SAW) method. Research on the domain of the problem mapping of flood-prone areas in the city of Semarang with the criteria of rainfall, topography, drainage, and land use succeeded in providing information on the level of flood vulnerability in each area by this method [16]. Other research to consider employee salary increases to increase company productivity also implements the SAW method and can produce a percentage increase in salary for each employee [15].

Some research on decision support systems using SWRL reasoning has also been done before. Decision Support System research that aims to select the manufacturing process shows that rule-based reasoning using SWRL allows being implemented in the decision making process [8]. Other Decision Support Systems to assist the labeling process of environmentally friendly products also show how a decision support system with an SWRL Reasoning-based mechanism can produce explanations and arguments by eco-labeling criteria [22]. The SWRL Reasoning technology in such ontology-based systems is proven to be used in decision support systems for various problem domains [17].

In this study, the SAW method is used for weighting criteria and determining alternatives on several criteria whereas, SWRL is used to make final decisions in the form of slices of alternative departments each of the following criteria with the value of each alternative.

3 The Approach

The idea is to improve the previous work, which considers the two-way approach [20], which has shown promising results. Unfortunately, it has not yet considered the talent interest, which is based on an assessment of psychology experts. Therefore, this work focus on using four aspects. K1 is Holland Talent Interest Test. K2 is Academic Value. K3 is Department Main Material, and K4 is Department Main College Activities. The other issue, users have the right to emphasize the preference on which group of calculations as that the users are the decision-maker. By considering this problem situation, the characteristic of SAW [15], which allows the subjectivity of users satisfied the solution of this decision problem. The calculation of K1 follows the rule in Holland Test. Meanwhile, the K2, K3, and K4 follow the method of SAW. In the end, to combine all is used SWRL.

3.1 The Combination of SAW and SWRL

At this stage, the SAW method and SWRL reasoning are implemented in the decision support system in this study. The step of

implementation and experiment are described in the pseudo-code as follows:

```

Input      : Score preference criteria, Data of Student (in group K1, K2, K3, K4)
Output     : The chosen alternative (A*)
Step      :
Begin
  Calculate the preference criteria
  Calculate of student's data
    Calculate data group K1
    Calculate the domain category
    Compose combination of category
    Calculate the weight of the combination
    Normalization of weight combination
    Conversion of university's major study
    The chosen alternative (A1)
  End
  Calculate data group K2
  SAW computation of K2
  The chosen alternative (A2)
  End
  Calculate data group K3
  SAW computation of K3
  The chosen alternative (A3)
  End
  Calculate data group K4
  SAW computation of K4
  The chosen alternative (A4)
  End
  SAW computation of A1, A2, A3, A4
  Obtaining a weighted score of A1, A2, A3, A4
  Processing reasoning of SWRL
  Obtaining A*
End
  
```

Implementation of the aptitude interest test approach using SWRL and SAW method consists of several stages. The first stage is entering the preferences of each criterion towards decision making based on the subjective preferences of prospective students. This value is then calculated into the criteria preference weight. After data is inputted, the Holland Test is calculated on K1 data and K2, K3, K4 data using the SAW method.

K1 data processing begins by calculating the combination weights for each of Holland's categories to get three dominant types called A, B, C. Then, three dominant categories are arranged into 6 level combination categories with combination weights being the total weight of the constituent categories added to the combination level number. These six levels are respectively; CBA, CAB, BCA, BAC, ACB, ABC. Furthermore, the combination of categories is converted into a department. The weight of the department is the result of the normalization of the category weight's combination. The conversion of category combinations into departments is based on knowledge in the Knowledge Base. The results of this conversion then become Alternative K1.

Furthermore, K2, K3, and K4 data are processed separately. Each SAW computation is performed in each criterion, which then generates the separated alternatives based on each criterion, namely alternative K2, K3, and alternative K4. Furthermore, all alternative values in Alternative K1, Alternative K2, Alternative

K3, and Alternative K4 are processed with SAW computation so that Alternative K1, K2, K3, and K4 are weighted using criteria preference weights.

The last step is determining the outcome of the decision. This stage uses SWRL reasoning that has been compiled in the ontology. This rule governs decision making in the form of Alternative slices K1, K2, K3, and K4 weighted with the final alternative value in the form of the sum of the weighted alternative values of each criterion.

3.2 Preparation of Ontology

The knowledge-Base is developed, which contains the basis of the knowledge system in the implementation of Holland Tests and decision making on department selection based on the results of Holland Tests and SWRL rules for final decision making by taking alternative slices obtained from each criterion. The built Knowledge-Base defines the characteristics of each of the Holland categories. Then, each value of type becomes a question in the Holland Test package and the suitability of the combination of the Holland category with alternative departments.

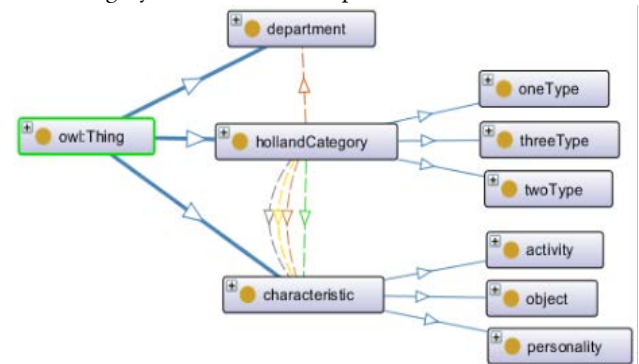


Figure 1. The schema of the ontology

In the SWRL rules that are arranged, each incoming data is associated with individual $?p$, where the data $?a$ is an alternative value of the criteria K1. b is an alternative value of the K2 criterion, c is an alternative value of the K3 criterion, d is an alternative to the criteria K4, as well as $?sum$ is the sum of the results of a , b , c , and d as the alternative value of the final decision. The example of SWRL rules developed to determine alternatives to the final decision can be seen as below :

The SWRL for Informatics department (IF)

```

person (?p),
has_career_informatics (?p, ?a),
has_academic_informatics (?p, ?b),
has_lecture_informatics (?p, ?c),
has_practicum_informatics (?p, ?d),
add (?sum, ?a, ?b, ?c, ?d)
has_final_informatics (?p, ?sum)
  
```

The SWRL is also preserved for all majors, which are calculated in the experiment of this work.

3.3 Determining of Criteria Preference Weight

At this stage, the criteria preference weight is calculated based on the criteria preference values that the user has inputted. Users input the value based on their preference. The process of calculating the criteria weights are shown in the following pseudo-code:

Input : value1, value2, value3, value4

Output : weights 1, weights 2, weights 3, weights 4

Step :

```
Function weight (value1, value2, value3, value4)
    total = value1 + value2 + value3 + value4;
    for (i = 1 to 4)
        weight.i = 1 / total * value.i;
    End
```

3.4 Alternatives Based on Each Criterion

At this stage, the calculation is done based on each criterion. The following are the results of calculating alternative values based on each criterion for respondents. For A1, there is a weight category of obtained combinations based on K1 such as ASE, AES, AES, and SAE. For A2, A3, and A4, there is a weight of department alternatives based on respectively K2, K3, and K4. The explanations of those departments are described in Table 1.

Table 1. Alternative Departments

Code	Name of Department
SI	Indonesian literature
PBK	Counseling guidance
TE	Electrical Engineering
IF	Informatics
DKV	Visual communication design
DI	Design interior
KD	Medical
MAT	Mathematics
PSI	Psychology
TS	Civil Engineering
TM	Mechanical Engineering
MJ	Management
PMAT	Mathematics education
BIO	Biology
TK	Chemical Engineering
PSD	primary teacher Education
TAR	Architecture
PAUD	Early childhood education programs

3.5 Decision on Department Selection

To decide the department selection results obtained from alternative slices of the four existing criteria with weights obtained

from the number of department alternative values on each criterion. The decision making was carried out using SWRL Reasoning. Examples of SWRL reasoning processes for alternative PBK on M1 data are as follows:

```
person (M1),
has_career_guidanceAndCounseling (M1, 0.224),
has_academic_guidanceAndCounseling (M1, 0.138),
has_lecture_guidanceAndCounseling (M1, 0.257),
has_practicum_guidanceAndCounseling (M1, 0.138),
add (? sum, 0.224, 0.138, 0.257, 0.138)
has_final_guidanceAndCounseling (M1, 0.757)
```

From the process of reasoning, in the final decision on the results obtained in the form of alternative courses, 0.757 PBK with alternative values. The rest values are SI 0.641, PSD 0.606, DI 0.585, and DKV 0.474. Therefore, the chosen alternative (A*) is the department PBK.

4 The Experiment

4.1 Data Collection

4.1.1 Collection of Holland Test Data. Holland's test data for this study is from the Indiana University - Purdue University Indianapolis website (<https://www.cco.purdue.edu/Students/CareerCompass>). This data contains the characteristics of the Holland category for aptitude interest tests and courses that are appropriate for the Holland category combinations.

The characteristics of the Holland category are grouped into four sections, namely Interest, Values, Personality, and Skill. These characteristics are used as questions in the test. Those questions are categorized into; Realistic, Investigative, Artistic, Social, Enterprise, and Conventional. Table 2 shows an example of a characteristic of the Realistic category.

Table 2. Characteristics of Realistic Category

Code	Part	Characteristics
R1	Interest	Building / Fixing
...	...	...
R7	Interest	Producing Clear Results
R8	Values	Clear Goals, Structure
...	...	...
R14	Values	Tradition, Craftsmanship
R15	Personality	Reliable, Dependable
...	...	...
R21	Personality	Action-Oriented
R22	Skill	Building / Fixing
...	...	...
R26	Skill	Self-Reliance

4.1.2 Data Collection of Department. The department data collection is done by interviewing with the Head of the Department or the Lecturer of Departments. The data obtained is data for criteria K2, K3, and K4. Data for the K2 criteria are ranked for each available option according to the conditions of the department. Then the data for K3 and K4 in the form of several answer choices

for each criterion. At this stage, data were collected from 18 departments with a total of 63 data for K3 and 52 data for K4. From the department data collection, data collected for K2, K3, and K4 criteria. Examples of K2 data for PAUD departments are as follows in Table 3, Table 4, and Table 5:

Table 3. PAUD K2 Criteria Data

Code	Subjects	Ranking
K2-1	Religious Education	1
K2-2	Civic Education	6
...	...	...
K2-14	Economy	14

Table 4. K3 Criteria Data

Code	Main Material	Department
K3-1	Indonesian	SI, PSD
K2-2	Creative Writing	SI
...	...	...
K3-63	Teaching and Learning Strategies for PAUD	PAUD

Table 5. K4 data

Code	Class Activity	Department
K4-1	Study of Literary Works	SI
K4-2	Philology Laboratory	SI
...	...	...
K4-52	Educational Games	PAUD

4.1.3 Respondent Data Collection. There are two classes of respondents in this study, namely college students/ alumni of the 2013-2016 and prospective students of the 2020 class. For the class data of students/alumni, they can already know and feel clear whether the chosen department is suitable or not. As for prospective students, they are the subjects aimed at the benefits of this study. Total data collected from 66 respondents, with 31 Student/Alumni and 35 Student Candidate. Respondent data is obtained by distributing online questionnaires.

In K1, the respondent answers the Holland test question. The respondent fills in the latest report card grades for 14 high school subjects in K2. In K3, the respondent responds to the preference of interest in main courses in the department. In K4, the respondents to choose the daily activities in each department. The respondents also input the preference interest of each criterion to determine the effect of it on each criterion. The criteria preferences of one criterion with other criteria can be the same or different. The answer choices along with the conversion of quantitative values for preference criteria and answers K1, K3, K4 are Not Important (0), Less Important (1), Important (2), and Very important (3). The answer option of K1 is Yes (1) or No (0). Meanwhile, the answer option of K3 and K4 are Highly Not Interest (0),

Not interested (1), Neutral (2), Interest (3), or Highly interested (4). Note: the number inside the bracket is the weight that will be used in the calculation.

4.2 Results

The evaluation is done by comparing the results of the system and the approval rating of the respondents. Respondents give an Agree / Disagree rating for an alternative order of departments produced by the system. From a total of 66 respondents, 64 respondents who gave an approval rating consisting of 33 prospective student respondents, and 31 student respondents. The results of the agreement level of the prospective student respondent group, the student respondent group, and the combination of the two groups of respondents are shown in Tables 6, 7, and 8.

Table 6. Level of Suitability of Prospective Students

Suitability	Amount	Percentage (%)
Agree	31	93.94%
Disagree	2	6.06%

Table 7. Student Compatibility Level

Suitability	amount	Percentage (%)
Agree	30	96.77%
Disagree	1	3.23%

Table 8. Overall Compliance Level of Respondents

Suitability	Amount	Percentage (%)
Agree	61	95.31%
Disagree	3	4.69%

Based on the evaluation results above, it was shown that the alternatives produced by the system in this study could be implemented in decision making in selected departments.

Several things have become the limitations of this study. Among them are limited data, including Holland data that was not obtained from the main test equipment sources and alternative data for the Sebelas Maret University's department, which only amounted to 18 alternatives and the limitations of the less complex alternative value merging methods. This alternative limitation has a significant impact on the respondent's approval response, where there is a group of prospective student respondents who tend departments that are not available in alternatives such as departments in the agriculture and social science faculties.

Another limitation is that there is no difference in the treatment applied to the two groups of respondents. Hence, it is assumed that these two groups can test the system with the same procedure. The ontology compiled in this study is also simple so that it has not been able to demonstrate the optimal use of SWRL.

5 Conclusions and Future Work

This work has been able to implement SWRL and SAW methods in the Department Selection Decision Support System with the Talent Interest Test Approach. In this study, the selection of departments according to talent interests. The test is based on Holland Tests, which is done directly in the system. The test results produce a combination of dominant categories on the subject, which then are converted into courses that are suitable for that combination. The result of this conversion becomes Alternative K1. Then the department alternative is determined based on other criteria separately, namely the criteria of Academic Ability, Interest in Lecture Material, and Interest in Lecture Activities using SAW computing to produce Alternative K2, Alternative K3, and Alternative K4. The alternative of each criterion is given a weight based on the criterion's preference weight. Then the slices of the four criteria are taken using the SWRL rule so that it produces the final decision. By calculating the level of conformity with the results of the department final decision obtained by the system based on the approval rating of 64 respondents, the concordance level was agreed (95.31%) and Disagree (4.69%).

Suggestions that can be considered for further assessment are as follows; improvements to the limitations in this study, including work with psychology practitioners to get Talent Interest Test data from the main reference source. Adding alternative departments data that is more diverse and from various faculties. Using a more complex alternative value merging methods. Giving different treatment to the two groups of respondents who tested the system so that more accurate evaluation results were obtained. Ontology improvements for more optimal use of SWRL; respondent data used should be more balanced and balanced for all alternative departments and increasing the level of confidence for each outcome decision.

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